

```
import pandas as pd
df = pd.read_csv('spam_dataset.csv')
print(df.head())
```

```
      v1                                     v2
0  ham                                     Hello how are you doing today
1  spam  Congratulations! You won a free lottery prize ...
2  ham           Can we meet tomorrow for project discussion
3  spam           Win cash now click this link immediately
4  ham           Please send me the assignment file
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

```
from google.colab import files
uploaded = files.upload()
```

spam_dataset.csv
spam_dataset.csv(text/csv) - 1919 bytes, last modified: 02/05/2026 - 100% done
 Saving spam_dataset.csv to spam_dataset (1).csv

```
df = pd.read_csv("spam_dataset.csv", encoding='latin-1')
```

```
print(df.head())
```

```
      v1                                     v2
0  ham                                     Hello how are you doing today
1  spam  Congratulations! You won a free lottery prize ...
2  ham           Can we meet tomorrow for project discussion
3  spam           Win cash now click this link immediately
4  ham           Please send me the assignment file
```

```
df = df[['v1', 'v2']]
df.columns = ['label', 'message']
```

```
df['label_num'] = df.label.map({'ham':0, 'spam':1})
```

```
X_train, X_test, y_train, y_test = train_test_split(
    df['message'],
    df['label_num'],
    test_size=0.2,
    random_state=42
)
```

```
vectorizer = TfidfVectorizer(stop_words='english')

X_train_tfidf = vectorizer.fit_transform(X_train)
X_test_tfidf = vectorizer.transform(X_test)
```

```
model = MultinomialNB()
model.fit(X_train_tfidf, y_train)
```

▼ MultinomialNB ⓘ ?
 MultinomialNB()

```
y_pred = model.predict(X_test_tfidf)
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

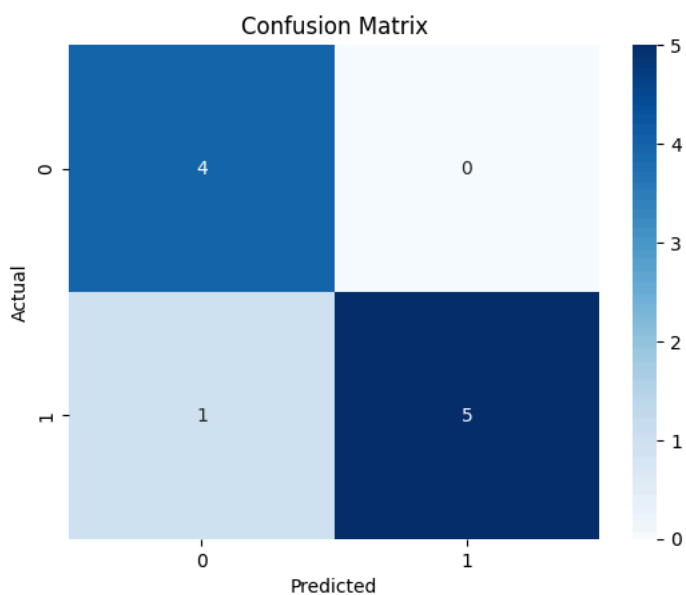
```
Accuracy: 0.9
```

```
print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
0	0.80	1.00	0.89	4
1	1.00	0.83	0.91	6
accuracy			0.90	10
macro avg	0.90	0.92	0.90	10
weighted avg	0.92	0.90	0.90	10

```
cm = confusion_matrix(y_test, y_pred)
```

```
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```



```
def predict_spam(msg):
    data = vectorizer.transform([msg])
    pred = model.predict(data)[0]

    if pred == 1:
        print("Spam Email")
    else:
        print("Not Spam Email")
```

```
predict_spam("Congratulations! You won 50000 rupees. Click now!")
```

```
Spam Email
```

